Gabriel MARIADASS | Paris, France

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Portfolio: mariadass-gabriel.github.io

SUMMARY

M.Sc. MVA at ENS Paris-Saclay and M.Eng. at École Nationale des Ponts et Chaussées, combining applied ML engineering at Dassault Systèmes and Banque de France with research in uncertainty-aware ML at NTU Singapore. Interested in a CIFRE PhD or an ML Engineering role in industry, startup or scale-up. Available from October 2026.

EDUCATION

ENS Paris-Saclay, France *2025 - 2026*
M.Sc. Mathematics, Vision, and Machine Learning (MVA)

- **Academic average:** 17.19/20 (with Highest Honors)
- **Key courses**: Deep Learning, Advanced Learning for Text and Graph Data (ALTEGRAD), Graphs in Machine Learning, Bayesian Machine Learning, Convex Optimization for Machine Learning, Computational Statistics, Machine Learning for Time Series

École Nationale des Ponts et Chaussées, France *2022 - 2026*
M.Eng. - Mathematical and Computer Engineering (IMI)
Final year completed at ENS Paris-Saclay (MVA)

Lycée Henri-IV, France *2020 - 2022*
*CPGE - PCSI/PC**

RESEARCH EXPERIENCE

LPSM, France & NTU, Singapore *04/2026 - 08/2026*
Research Intern - Possibilistic Variational Inference
Supervisors: Badr-Eddine Chérief-Abdellatif (CNRS, LPSM) and Jérémie Houssineau (NTU)

- Reproduced published image-classification results (LeNet-5, Fashion-MNIST) to build a validated baseline for new methods
- Identified why the model's uncertainty estimates were unreliable on four test cases and traced it to the choice of model family
- Designed learned uncertainty structures across four model families, improving accuracy, NLL, and calibration
- Built on an existing SAM connection to derive an adaptive, automatically learned perturbation radius

PROFESSIONAL EXPERIENCE

Dassault Systèmes, France *01/2025 - 07/2025*
ML Engineer - Deep Learning & GNN (Molecular Retrosynthesis)

- Migrated GNN retrosynthesis pipeline from DGL to PyTorch Geometric, cutting CPU inference time by 38%
- Evaluated Bayesian hyperparameter optimization (+2 points accuracy) but rejected it for a 2.8x heavier model
- Explored quantization (dynamic/static) for inference speedup and refactored the preprocessing pipeline for maintainability

Banque de France, France *09/2024 - 12/2024*
Data Scientist - On-site Credit Institution Control Delegation

- Built neural network credit-scoring models with AUC 0.83 on 700K loan contracts, raising Gini +0.14 vs. internal IRB models
- Led PD/LGD/EAD risk-parameter analysis and time-series studies for a live ECB-mandated audit of a major French bank

TECHNICAL SKILLS

- **Programming**: Python, SQL
- **ML & Deep Learning**: PyTorch, scikit-learn, XGBoost/LightGBM, NumPy, Pandas, tabular modeling, time-series forecasting, GNNs (PyTorch Geometric, DGL), text embeddings and retrieval, LoRA fine-tuning, Bayesian optimization
- **ML Engineering & Deployment** (see price-api in portfolio): Docker, FastAPI, MLflow, CI/CD (GitHub Actions), Cloud (GCP), pytest, Git, model quantization
- **Mathematics**: Convex Optimization, Bayesian Learning, Variational Inference, Time Series, Information Theory, MCMC, PAC-Bayes, Possibility Theory

LANGUAGES & INTERESTS

- **Languages**: French (native), English (C1), Spanish (B1)

- **Interests**: Muay Thai / Kickboxing, Running, Weightlifting, Nutrition, AI Tech Watch, Tech Entrepreneurship, Reading, Travel